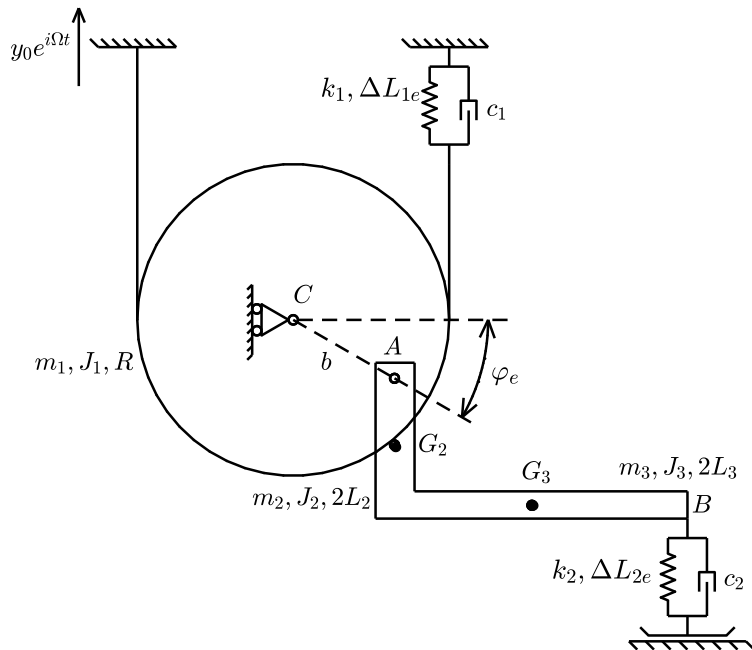


Dynamics of Mechanical Systems

November 04th, 2022



$m_1 = 5 \text{ kg}$	$k_1 = 2000 \text{ N/m}$
$m_2 = 5 \text{ kg}$	$k_2 = 4000 \text{ N/m}$
$m_3 = 5 \text{ kg}$	$\Delta L_{2e} = -0.0061 \text{ m}$
$J_1 = 0.6 \text{ kgm}^2$	$c_1 = 4 \text{ Ns/m}$
$J_2 = 0.4 \text{ kgm}^2$	$c_2 = 0.5 \text{ Ns/m}$
$J_3 = 0.4 \text{ kgm}^2$	$\varphi_e = \pi/6$
$R = 0.5 \text{ m}$	
$b = 0.4 \text{ m}$	
$L_2 = 0.50 \text{ m}$	
$L_3 = 0.50 \text{ m}$	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk is connected to the ground through a cart in its center C and the L-shaped rigid body is connected with a hinge to the disk in the eccentric point A. According to a different set of symbols used in this course, $\varphi_e = \varphi_{st}$, $\Delta L_{1e} = \Delta L_{1st}$ and $\Delta L_{2e} = \Delta L_{2st}$.

The students are requested to:

0. Write on top of each page their surname, name, Person Code and **SNxxxx code** (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the horizontal component of the displacement of the center of mass G_3 and the vertical component of the displacement of the center of mass G_2 . Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxxx are the last four digits of your Person Code (Codice Persona) (e.g. for Bruni Stefano 12345678: BS56781.fig and BS56782.fig)
4. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the vertical component of the constraint force in the hinge A. Save the graphical result as SNxxxx3.fig.
5. Compute the static preload ΔL_{1e} of spring k_1 .

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one.

You must upload one single archive file, which should contain your MATLAB script and all the .fig files you want to deliver as part of your solution.

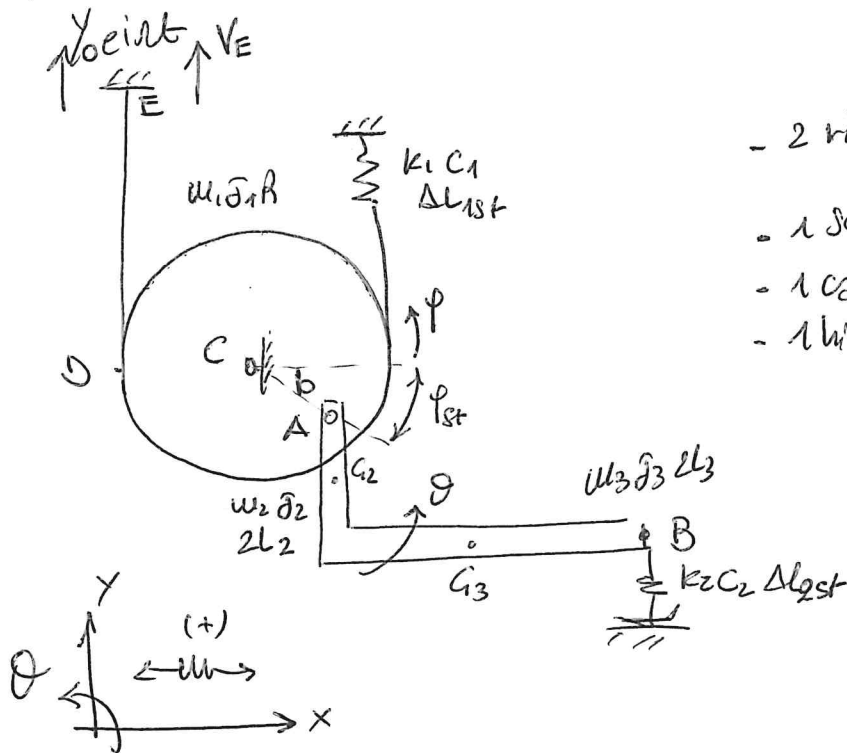
- Accepted formats for the archive files are: .7z .rar .zip;
- The file must be named SNxxxx with “S” the first letter in your surname, “N” the first letter in your name (in case of multiple surnames/names just consider the first one), “xxxx” the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
- The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test 4th_Nov_2022_L01, Test 4th_Nov_2022_L03 or Test 4th_Nov_2022_L09 depending on the room where you are having the test** (make sure you choose the correct folder!);
- the limit time to upload the assignment will be announced during the written exam.

Please be aware that your answers to the test will not be considered in case:

- the information stated at point 0 above is missing;
- your archive file is uploaded in the wrong assignment folder
- the name of the archive file is different from the SNxxxx code as explained above;

DYNAMICS OF MECHANICAL SYSTEMS

Written Test November 4th 2022



- 2 rigid bodies $\rightarrow$ 6 dof. without constraints.

- 1 spring ED $\rightarrow$ -1 dof

- 1 cart C $\rightarrow$ -1 dof

- 1 knife A $\rightarrow$ -2 dof

$\downarrow$
2 dof.

$$\underline{x}_F = \begin{Bmatrix} \phi \\ \theta \end{Bmatrix} \rightarrow \underline{x} = \begin{Bmatrix} \phi \\ \theta \\ y \end{Bmatrix} \quad \underline{x}_{st} = \begin{Bmatrix} \phi_{st} \\ \theta \\ 0 \end{Bmatrix}$$

$$\underline{x}_c = \begin{Bmatrix} y \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 \dot{y}_c^2 + \frac{1}{2} J_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{x}_{c2}^2 + \frac{1}{2} m_2 \dot{y}_{c2}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_{c3}^2 + \frac{1}{2} m_3 \dot{y}_{c3}^2 + \frac{1}{2} J_3 \dot{\theta}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 c_i \Delta L_i^2; \quad V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta L_i^2; \quad V_g = \sum_{i=1}^3 m_i g h_{ci}$$

$$\delta W = \vec{V}_E \times \delta \vec{y} = V \delta y$$

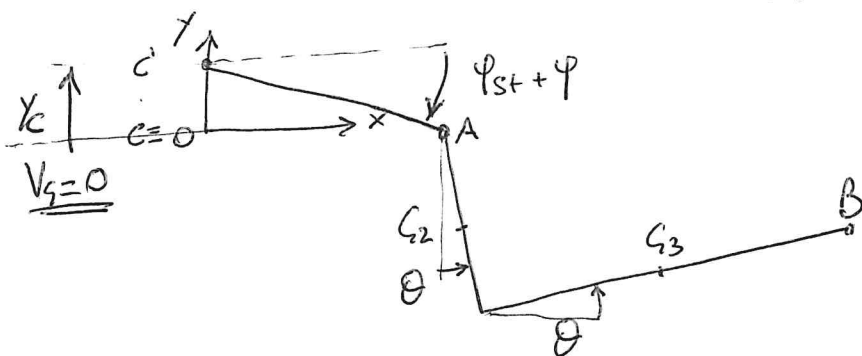
$$[m_{PH}] = \text{diag}(m_1, J_1, m_2, m_2, J_2, m_3, m_3, J_3); \quad \underline{y}_{PH} = \{ y_c, \phi, x_{c2}, y_{c2}, \theta, x_{c3}, y_{c3}, \theta \}^T$$

$$[c_{PH}] = \text{diag}(c_1, c_2); \quad [k_{PH}] = \text{diag}(k_1, k_2); \quad \underline{y}_{ke} = \underline{y}_e = \{ \Delta L_1, \Delta L_2 \}^T$$

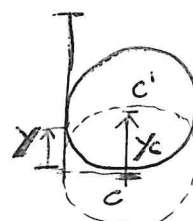
$$\underline{f}_{PH} = \{ V_E \}; \quad \underline{y}_f = \{ y \}$$

• non-linear kinematic relationships

according to the sign convention, $\phi_{st} = -\frac{\pi}{6}$



$$y_c = y + \phi R$$



$$Y_c = Y + PR$$

$$x_A = b \cos(\psi_{st} + \varphi)$$

$$Y_A = Y + PR + b \sin(\psi_{st} + \varphi)$$

$$x_{s2} = b \cos(\psi_{st} + \varphi) + L_2 \sin \theta$$

$$Y_{s2} = Y + PR + b \sin(\psi_{st} + \varphi) - L_2 \cos \theta$$

$$x_{s3} = b \cos(\psi_{st} + \varphi) + 2L_2 \sin \theta + L_3 \cos \theta$$

$$Y_{s3} = Y + PR + b \sin(\psi_{st} + \varphi) - 2L_2 \cos \theta + L_3 \sin \theta$$

$$Y_B = Y + PR + b \sin(\psi_{st} + \varphi) - 2L_2 \cos \theta + 2L_3 \sin \theta$$

$$\Delta L_1 = -Y - 2PR$$

$$\Delta L_2 = Y_B - Y_{B,st}$$

$$\frac{\partial x_{s2}}{\partial \varphi} = -b \sin(\psi_{st} + \varphi) ; \frac{\partial x_{s2}}{\partial \theta} = L_2 \cos \theta$$

$$\frac{\partial Y_{s2}}{\partial \varphi} = R + b \cos(\psi_{st} + \varphi) ; \frac{\partial Y_{s2}}{\partial \theta} = L_2 \sin \theta ; \frac{\partial Y_{s2}}{\partial Y} = 1$$

$$\frac{\partial x_{s3}}{\partial \varphi} = -b \sin(\psi_{st} + \varphi) ; \frac{\partial x_{s3}}{\partial \theta} = 2L_2 \cos \theta - L_3 \sin \theta$$

$$\frac{\partial Y_{s3}}{\partial \varphi} = R + b \cos(\psi_{st} + \varphi) ; \frac{\partial Y_{s3}}{\partial \theta} = 2L_2 \sin \theta + L_3 \cos \theta ; \frac{\partial Y_{s3}}{\partial Y} = 1$$

$$\frac{\partial \Delta L_1}{\partial \varphi} = -2R ; \frac{\partial \Delta L_1}{\partial Y} = -1$$

$$\frac{\partial \Delta L_2}{\partial \varphi} = R + b \cos(\psi_{st} + \varphi) ; \frac{\partial \Delta L_2}{\partial \theta} = 2L_2 \sin \theta + 2L_3 \cos \theta ; \frac{\partial \Delta L_2}{\partial Y} = 1$$

$$\Delta u = \begin{bmatrix} R & 0 & 1 \\ 1 & 0 & 0 \\ -b \sin \psi_{st} & L_2 & 0 \\ R + b \cos \psi_{st} & 0 & 1 \\ 0 & 1 & 0 \\ -b \sin \psi_{st} & 2L_2 & 0 \\ R + b \cos \psi_{st} & L_3 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \Delta k = \Delta c = \begin{bmatrix} -2R & 0 & -1 \\ R + b \cos \psi_{st} & 2L_3 & 1 \end{bmatrix}$$

$$\Delta f = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 \Delta L_2}{\partial \varphi^2} = -b \sec(\varphi_{st} + \varphi) \\ \frac{\partial^2 \Delta L_2}{\partial \theta^2} = 2L_2 \cos \theta - 2L_3 \sec \theta \end{cases} \rightarrow [k_{el II, 2}] = k_2 \Delta L_{2st} \begin{bmatrix} -b \sec \varphi_{st} & 0 & 0 \\ 0 & 2L_2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$h_{s1} = \gamma_c = \gamma + \varphi R \rightarrow$ linear function $\rightarrow [k_{s1}] = 0$

$\Delta h_1 = -\gamma - 2R\varphi \rightarrow$ " " $\rightarrow [k_{el II, 1}] = 0$

$h_{s2} = \gamma_{s2}$; $h_{s3} = \gamma_{s3}$

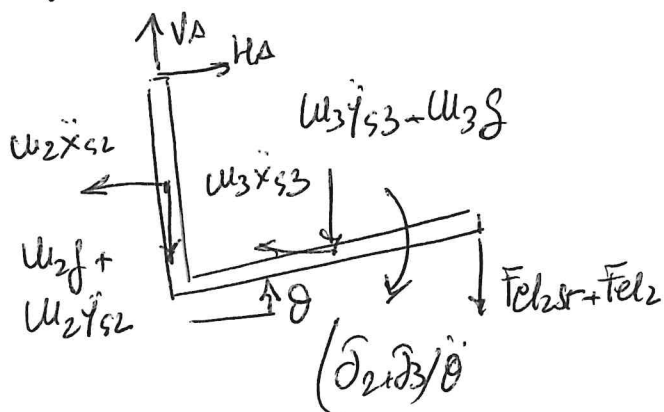
$$\begin{cases} \frac{\partial^2 h_{s2}}{\partial \varphi^2} = -b \sec(\varphi_{st} + \varphi) \\ \frac{\partial^2 h_{s2}}{\partial \theta^2} = L_2 \cos \theta \end{cases} \rightarrow [k_{s2}] = u_2 g \begin{bmatrix} -b \sec \varphi_{st} & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 h_{s3}}{\partial \varphi^2} = -b \sec(\varphi_{st} + \varphi) \\ \frac{\partial^2 h_{s3}}{\partial \theta^2} = 2L_2 \cos \theta - L_3 \sec \theta \end{cases} \rightarrow [k_{s3}] = u_3 g \begin{bmatrix} -b \sec \varphi_{st} & 0 & 0 \\ 0 & 2L_2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- matrix partition :

$$\begin{bmatrix} [M_{ff}] & [M_{fc}] \\ [M_{cf}] & [M_{cc}] \end{bmatrix} \begin{matrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{matrix}$$

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$$\sum F_y = 0 :$$

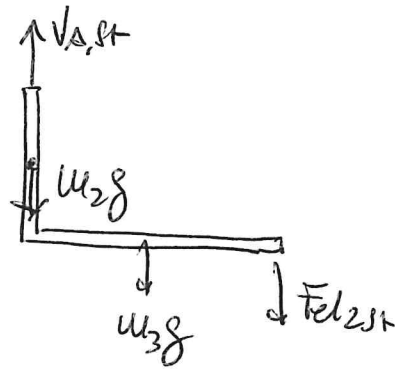
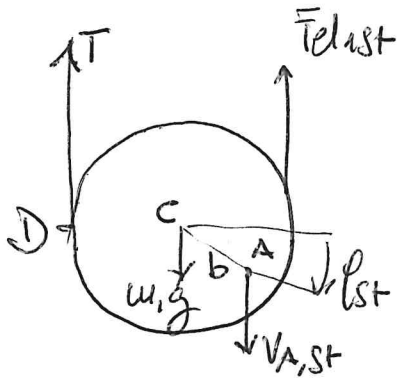
$$V_A - u_2 g - u_3 g - F_{c2st} - u_2 \ddot{\gamma}_{s2} + u_3 \ddot{\gamma}_{s3} - F_{c2} = 0$$

↓

$$V_{A,st} = u_2 g + u_3 g + F_{c2st}$$

$$V_{A,dyn} = u_2 \ddot{\gamma}_{s2} + u_3 \ddot{\gamma}_{s3} + F_{c2}$$

question nr 5 :



$$\sum F_y = 0 \rightarrow V_{A,st} = w_{2g} + w_{3g} + F_{el2,st}$$

$$\sum M_D^{disk} = 0: F_{el1,st} 2R - w_{1g} R - V_{A,st} (R + b \cos \psi_{1st}) = 0$$

$$F_{el1,st} = \frac{w_{1g} R + V_{A,st} (R + b \cos \psi_{1st})}{2R}$$

$$\Delta l_{1st} = \frac{F_{el1,st}}{k_1}$$